Neural Network for Automatic Target Recognition Over a Wide Range of Poses

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Abstract

An extremely simple neural network, equivalent to a linear discriminant, is applied to the recognition of three target classes and their orientation. Probability of detection and probability of identification of over 98% is achieved. Probability of false alarm, given a non-target region of the image with structure the size of the targets, is 11%.

Three image analysis techniques, all new in this application, are used to abstract critical information from an image for input to the neural network. A bandpass technique is used to find areas in an image with components the size of the targets (points of interest). A second bandpass is used to find structural details to identify the target, estimate azimuth and elevation, and to find the center of view of the target. A log-polar representation of the target, using only 65 pixels, is used to convey the image to the neural network. The combination of these three techniques gives ATR performance comparable to the results of careful human inspection of these images.

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Introduction

The development of Recursive Error Minimization (REM) equations for neural network learning has continued for several years at Martin Marietta Astronautics Group [1, 2, 3, 4]. Recently, the concept of the generation of an infinite training set from a small number of exemplars was used with the REM equations to show that handwritten digits could be recognized by a neural network with only sixteen exemplars of each digit [5]. This work uses the same approach to identify targets and to estimate the pose of targets in one-, two-, or three-dimensional images. The application here is to two-dimensional visual images. The approach applies equally to radar range images (one, two, or three mode), infrared (FLIR) or Synthetic Aperture Radar (SAR) images, sonar, and 3-D radar or laser range images.

Problem

One of the most difficult aspects of Automatic Target Recognition (ATR) is that a target, whether viewed in one dimension (radar range function), two dimensions (visual, infrared, SAR), or three dimensions (3-D radar or laser range), changes appearance drastically with pose. This problem can be approached in several ways. For the target as a whole, a reference library for many poses can be collected and the unknown signature compared with the library (as with correlation techniques). However, the reference library can be quite large. A second approach is to look for features (generally less sensitive to pose), use a model-based reasoning approach to form hypotheses on identification and pose, and then to look for additional features to prove or disprove the hypotheses. Model-based reasoning is powerful, but may require massive computational capacity.

A third approach, used in this work, is to use a neural network for target recognition. In principle, a neural network can learn features which are independent of pose, and generalize to new poses. In most previous work, the neural network learned to recognize the targets in the training set, but performance on a test set, particularly with significant translation or different poses, was much less satisfactory. This work demonstrates excellent performance on visual images of targets (maximum target dimension of 30 pixels) using very simple techniques for segmentation and preprocessing of the information given to the neural network. Further, the required neural network uses only a few hundred connections and no hidden nodes to identify targets and estimate pose.

Finding Areas of Interest in an Image

A conventional approach to the ATR problem takes the search object as given, and tries to find it in the scene. Correlation-based techniques are frequently used. If the object of search is the whole target, a large library of different poses is required. If the object of search is some feature of the target, a much smaller library is required, but model-based reasoning, or its equivalent, is required to determine if the features and their relationships truly define the target to acceptable probability. All such processes are very computation intensive, although optical techniques or parallel processing can make them practical.

If human vision is examined, a much different picture emerges. Attention is focused on a series of "interesting areas." Investigators have shown that a plausible analogue to the biological processing can be obtained by transforming a scene with Gabor functions. Local maximums in the transformed scene correspond to these "interesting areas." Gabor functions are wavelet functions. That is, they are effectively non-zero only over a defined distance. Further, they are edge detectors, so the effect is to find edges of a particular length at a specific orientation.

Such transforms can be easily approximated optically, but for ATR, a single function would be much more convenient. If a series of Gabor transforms at all orientations are summed (before detection), the result is approximately a Difference of Gaussians (DOG) transform, which is the classic transform related to vision. The DOG gives the end result of blurring an image with Gaussian blurs of two different lengths, and subtracting. This transform is also very simple optically. The scene is Fourier transformed with lenses. A block is inserted in the Fourier plane which passes a band of spatial frequencies (annular slot). A second Fourier transform is performed with lenses, and the result is detected. Note that detection implies the squaring of the magnitudes of the transformed scene.

Since this work is performed on a PC (IBM AT), computations are minimized by substituting first-order blurring for Gaussian blurring.

The DOG transform, its optical equivalent, and the approximation used in this work all have the characteristic that they emphasize the presence of features of a specific size range. To find "interesting" areas in these images, first-order blurs with characteristic lengths of 5 and 9 pixels were differenced and squared. The two-dimensional transform function equivalent to this process has a positive central region about 20 pixels across, about the size of the targets of interest. The difference of two blurs of an image must have a mean value of zero, so the mean of the detected image is just the variance (σ^2) of the magnitude image. Thus a simple method for determining interest areas is to create a binary image by thresholding the detected image at five times its mean ($\sqrt{5}\,\sigma$ of the magnitude image). The centroid of each of the resulting blobs is then a point of interest, and a 40×40-pixel tile is abstracted for subsequent analysis.

The purpose of this work required a large number of non-target tiles to train the neural network, so the blur lengths and threshold were chosen to give a large number of non-target tiles (false alarms). For actual use, parameters would be selected to minimize the number of false alarms, subject to the requirement that all targets should be detected. Ten terrain board images were taken. The image with the most false alarms was reserved for separate test. The false alarm tiles from the other nine images (80 tiles in all) were used for training. Figure 1 presents the tenth image, with centers of interest indicated by symbols. The symbols indicate whether the point of interest was a target or false alarm, with a triangle marking false alarms.

Processing of Image Tiles

With an area of interest selected, emphasis shifts to the features of that area, features whose configuration will identify the target, or indicate that the tile does not contain a target. The same processing used to select the tiles is repeated, but now with first-order blur lengths of 1.5 and 3 pixels. The equivalent transform function for this process has a positive central region about six pixels across.

Since the significance of the detected image is in the features, the brighter pixels, the image is normalized with the ratio of the mean square of the detected image to the mean value. The effect is to normalize the brighter pixels to a common range, since low-value pixels have a minimal effect on this ratio (fourth moment / second moment).

Figure 2a shows one of the training tiles, an armored personnel carrier viewed at 90° azimuth and 45° elevation. Figure 2b shows the same tile after transform, detection, and normalization. It is important to realize that Figure 2b is unaffected by *any* linear transform of brightness of Figure 2a. Even a change from a positive to a negative image has no effect. Most monotonic transforms, such as limiting, will have minimal effect on Figure 2b. As might be expected, the effect of time of day on infrared (FLIR) images is almost eliminated by this processing. Note, however, that in actual computation, quantization effects must always be considered. If a brightness transform on a quantized image reduces the number of states of the image, information is destroyed, and Figure 2b will be affected. Figure 2b contains the information necessary for recognition and estimation of pose, but in itself is completely inappropriate for input to a neural network. First, a neural network has no inherent sense of neighborhood. Shifting a pixel value to an adjacent pixel is no different than shifting it to the other side of the image. The result is that learning to recognize a target in different positions and rotations is very difficult. Second, Figure 2b has 1 600 pixels. A neural network with 1 600 input pixels would have a very large number of connections. Since recognition time is proportional to the number of connections, and learning time proportional to the square of the number of connections, practical applications require the minimization of the number of connections.

Returning to the examination of human vision, consider the distribution of sensors in the retina. In the fovea, sensors are very densely packed. As the distance from the fovea increases, sensor density drops dramatically. A simple approximation which has this characteristic is a log-polar coordinate system. Another characteristic of human vision is that already used in segmentation; that is, vision fixates on a point of interest. The origin of the log-polar coordinate system must be defined.

The technique used here to define the center of the coordinate system was very simple. Figure 2b was thresholded to give a binary image. The centroid of this binary image was taken as the origin of the log-polar coordinate system. This process was quite satisfactory for the images used in training and statistical tests, but was not as satisfactory for the cluttered terrain images. In retrospect, it is clear that the centering procedure should be guided more by vision considerations than by computer convenience.

With the origin determined, the size and number of the log-polar pixels must be chosen. The radius of the outer edge was chosen to be 13 pixels. The rationale was to pick a radius which included most of a target and very little of the surround. The appropriate radius is, of course, a function of target size, sensor resolution, and range. Four rings of pixels were used, with outer to inner radius ratios of $2^{3/4}$, so that the foveal spot radius was 1/8 of the outer radius. For tasks in which high resolution in the center is important, the foveal spot would be made proportionally smaller. Sixteen rays are used, spaced at 22.5°, outside the foveal spot. This defines the areas of 64 ring-sector pixels, plus the foveal spot, for a total of 65 inputs to the neural network. To find the values of the log-polar pixels, this is simply laid over Figure 2b with the appropriate center. All the rectangular pixels with centers inside a log-polar pixel are summed to get the value of the log-polar pixel. For input to the neural network, a constant average brightness is obtained by normalizing each log-polar pixel with the sum of all the log-polar pixels.

In order to give a visual image of this process, the log-polar mask was placed on Figure 2b, and each rectangular pixel within a log-polar pixel was replaced with the average value of all the rectangular pixels within that log-polar pixel. This resulted in Figure 2c. This process gives an accurate visual representation of the information given to the neural network. It is difficult to understand how such a small amount of information can be quite adequate to identify class and pose of three different targets.

Training the Neural Network

The first step in training a neural network is to define the desired outputs. For identification, this is very simple. For a four-class problem (tank, armored personnel carrier, rocket launcher, and non-target), four output nodes are used. If the input is a tank, the output node corresponding to tank should be +1, and the other three output nodes should be -1.

The desired output of a neural network for pose is not so simple. In this case, pose is defined by two angles, the elevation from horizontal of the line of sight from the target to the observer, and azimuth, the orientation of the target with respect to the observer. In this case, functional relationships are required, and the desired output cannot be simply ±1, but must assume specific values as a function of the pose of the training example. If angle values from 0° to 359.9° are simply mapped to the range of output node values (-1 to +1), the consequence is that adjacent poses (0° and 359.9°) have desired outputs of +1 and -1. A neural network has a very difficult task responding to such a discontinuity in desired output. A periodic function of angle does not have such a discontinuity. However, any periodic function of angle must be ambiguous; that is, at least two angles result in the same output.

With this in mind, an unconventional technique was devised. Two output nodes were used to represent each angle. The desired output for one node was the sine of the angle and the desired output for the other node was the cosine of the angle. This matches the range of an output node (-1 to +1). The angle produced by the neural network in response to test images is calculated with an arctangent function of two arguments. Further, a consistency check of the neural network's determination of an angle can be made by summing the squares of the two outputs (range from 0 to 2):

$$c = 1 - [1 - (\sin^2 + \cos^2)]^2$$

The consistency parameter, c, varies from zero (inconsistent) to one (consistent).

Our last paper [5] showed that "overtraining," the tendency of a neural network to learn training examples so specifically that it does not do well on independent test examples, can be nearly eliminated by appropriate generation of an infinite set of training examples from a small set of training exemplars. In that paper the problem was the recognition of handwritten digits, and the training examples were obtained with variations in size, orientation, and two translation dimensions of the exemplars.

In this application, variations in pose must be added. An exact calculation of such variation would require models of the targets and of the sensor systems, and is impractical for this purpose. The procedure we devised was to consider only the effect of perturbations of pose on the size of the target. Further, the only geometric feature of the target used in the calculation was an estimate of the vertical aspect ratio (height / $\sqrt{\text{length} \times \text{width}}$). Although this calculation is very crude, a neural network trained on exemplars spaced every 15° in both azimuth and elevation recognized about 97% of the other target images correctly. Figure 3 presents these images for all three targets. The difficulty of identification in some poses and the difficulty of pose estimation is very evident from examination of these images. The results presented here used exemplars spaced every 5°, so that about 99% of the target images are recognized correctly.

The generalization of the training exemplars to obtain the training examples occurs in six dimensions for this application: size, orientation, two translation dimensions, azimuth, and elevation. This seems very complex, but all the variations are combined to give a single interpolation for each log-polar pixel for each training example.

One of the advantages of the use of REM equations for neural network training is that an estimate of the contribution of each connection to the minimization of error is available. This estimate allows unnecessary connections to be removed from, and incorrectly-removed connections to be added to the network. This process is called REM Thinning [3]. It eliminates unnecessary connections, minimizing both the complexity of the neural network and the learning time.

One of the significant characteristics observed in the training for this application was that a reasonable threshold value for REM Thinning removes *all* hidden nodes. The preprocessing, bandpass and the log-polar pixels, has transformed the information in the image into a format in which weighted linear sums of the rectangular pixels, passed through a sigmoid operation, give excellent identification and estimate of pose. This is a dramatic illustration of the vital role of preprocessing in the practical application of a neural network.

Results

The data collected for this application consisted of 259 uncluttered images of each of the three targets and ten images of targets in a cluttered environment (terrain consisting of hills, bushes, trees, grass, and shadows). The neural network was trained on four classes, the three target classes and a non-target class which consists of points of interest (terrain images with structural features similar in size to the targets) which are not targets. Even with the parameters of the selection of points of interest set to give a large number of non-target points, only 110 non-target points could be found in the ten terrain images. Reserving the image with the most non-target points (30) for later test, only 80 non-target images were available for training. This reduced the ability of the neural network to identify this class. Further, all these images were used as training exemplars, so the only independent test of the non-target class is the application to the single terrain image which was reserved.

Since the neural network is trained on an infinite set of examples generated from the training exemplars, a test against the training exemplars is a good estimate of the results which would be obtained from a test against the test exemplars. The total target set consists of images every 2.5° in azimuth from 0° to 90°, and every 5° in elevation from 15° to 45°. The training exemplars were chosen to be images every 5° in azimuth from 0° to 90°, giving 133 images for each target, plus the 80 non-target images. The test exemplars were the remaining target images, 126 for each target.

Performance of the neural network against the training exemplars is given in Table I. The probability of detection and probability of identification were equal at 99.8%, while the probability of false alarm, given the selection of a non-target as a point of interest, was 11.2%. Note that the actual false alarm rate per unit area depends also on the probability that the terrain has structural features about the size of the targets, and on the parameters used to select points of interest. RMS azimuth error was 8.3° and RMS elevation error was 6.6°.

Table I Test of Training Exemplars							
	Neural Network Identification				Probabilities		
	Tank	APC	RL	Non-Target	Prob. Detect.	Prob. Ident.	Prob. False Alarm
Tank	132 / 99.2%	0 / 0%	0 / 0%	1 / 0.8%	99.2%	99.2%	
APC	0 / 0%	133 / 100%	0 / 0%	0 / 0%	100%	100%	
RL	0 / 0%	0 / 0%	133 / 100%	0 / 0%	100%	100%	
N-T	3 / 3.7%	4 / 5.0%	2 / 2.5%	71 / 88.8%			11.2%
Overall probability of detection = 99.8% probability of identification = 99.8% probability of false alarm, given a non-target point of interest = 11.2%							

Performance of the neural network against the test exemplars is given in Table II. Probability of detection was 99.5% and probability of identification was 98.9%, slightly worse than against the training exemplars. RMS azimuth error was 7.5° and RMS elevation error was 6.5°, slightly better than against the training exemplars.

Table II
Test of Test Exemplars

	Neural Network Identification				Probabilities	
	Tank	APC	RL	Non- Target	Prob. Detect.	Prob. Ident.
Tank	124 98.4%	0 0%	0 0%	2 1.6%	98.4%	98.4%
APC	0 0%	125 99.2%	1 0.8%	0 0%	100%	99.2%
RL	1 0.8%	0 0%	125 99.2%	0 0%	100%	99.2%

Overall
probability of detection = 99.5%
probability of identification = 98.9%

Finally, the neural network was applied to the terrain image shown in Figure 1. All three targets were correctly identified, although a manual adjustment of the origin of the log-polar representation was required for the tile with the tank image. This reinforces the observation that the origin determination should be more closely related to vision characteristics. The actual target tiles used are presented in Figure 4a. Although 30 non-target points of interest were found in Figure 1, they resulted in only 27 unique tiles being abstracted from the image. Three of these 27 non-target tiles were identified as targets (11.1%), almost identical to the results for the training exemplars (11.2%). Figure 4b shows the non-target points of interest incorrectly identified as targets. Tiles #20 and #22 are identical. Figure 4c shows the non-target points of interest correctly identified as non-targets. Tiles #27, #28, and #30 are identical. Thus the performance against one terrain image was quite satisfactory.

Conclusions

This work has demonstrated extraordinary ATR performance with a simple neural network. The key features are a new segmentation technique (points of interest with a low-spatial-frequency bandpass), new processing of image data for input to the neural network (high-spatial-frequency bandpass and use of a centered log-polar coordinate system), and a new method for construction of an infinite set of training examples from a limited set of training exemplars.

The parameters for determination of points of interest and the technique for centering the origin of the log-polar coordinate system require additional improvement. A new data set with poses of a full hemisphere of observation and a large set of terrain images with and without targets is desirable for a complete evaluation of these techniques.

Further data from other sensors (one-, two-, and three-dimensional images) can then be secured to demonstrate performance over a wide range of situations.

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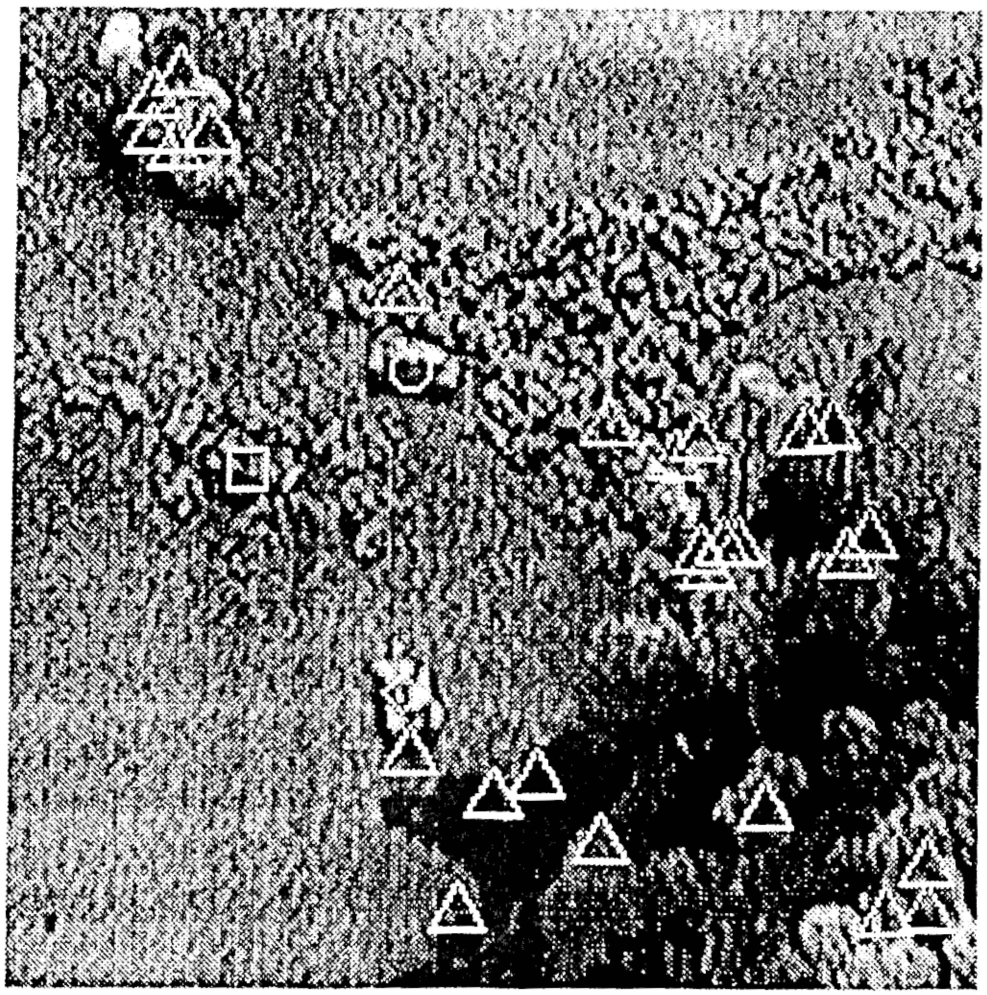
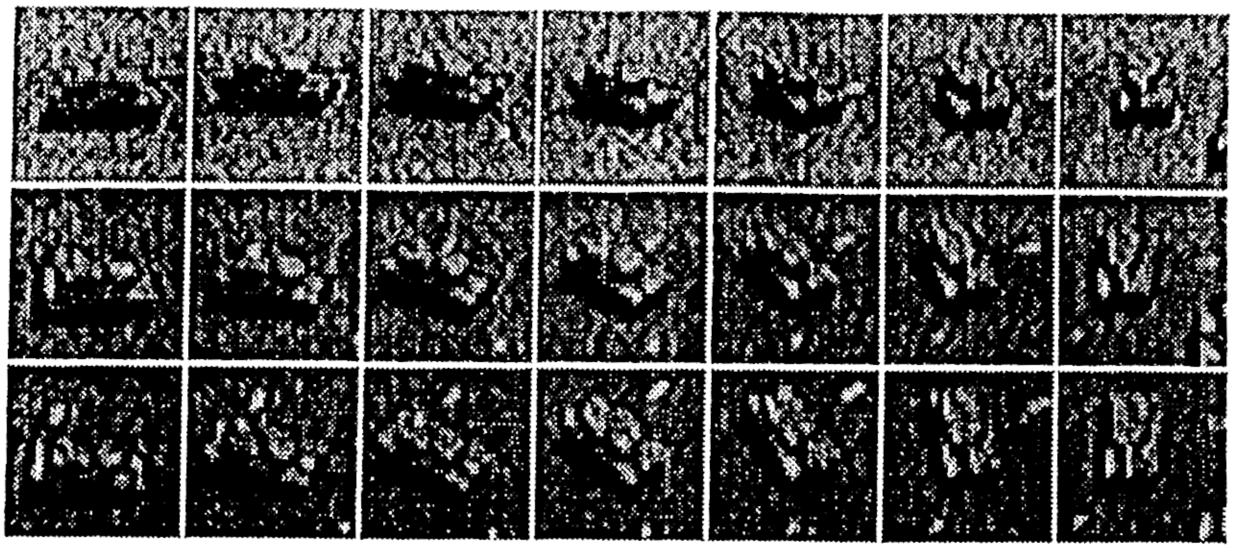


Figure 1. A terrain board image with points of interest marked. The tank is marked with a circle, the armored personnel carrier with a square, and the rocket launcher with a diamond. Non-target points of interest are marked with triangles

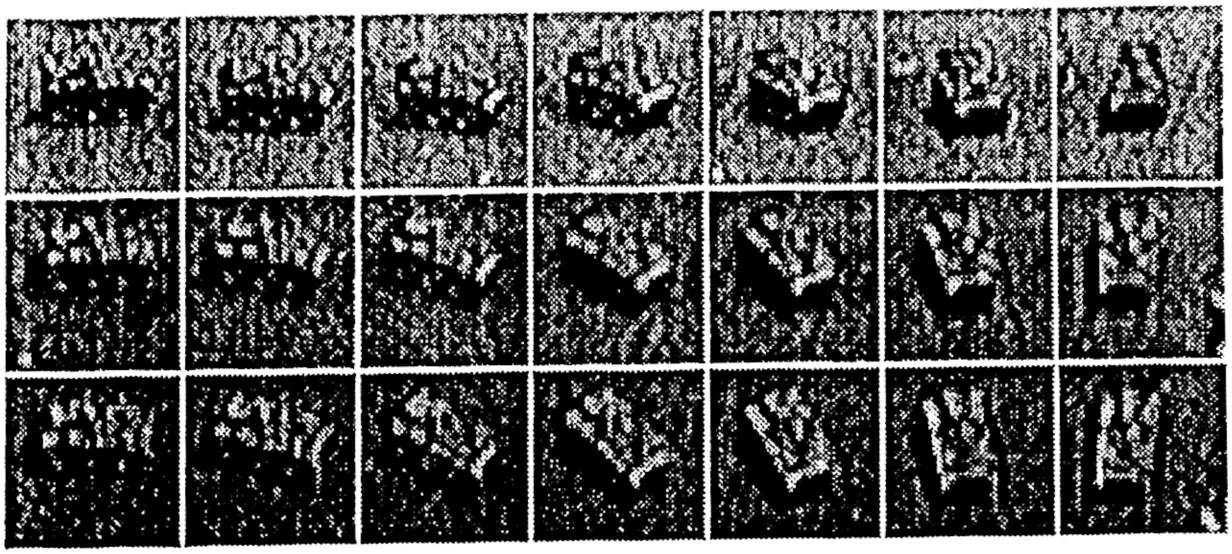
a. Brightness Image   b. Bandpass of a.   c. Log-polar version of b.

Figure 2. A training image of an armored personnel carrier at 90° azimuth and 45° elevation



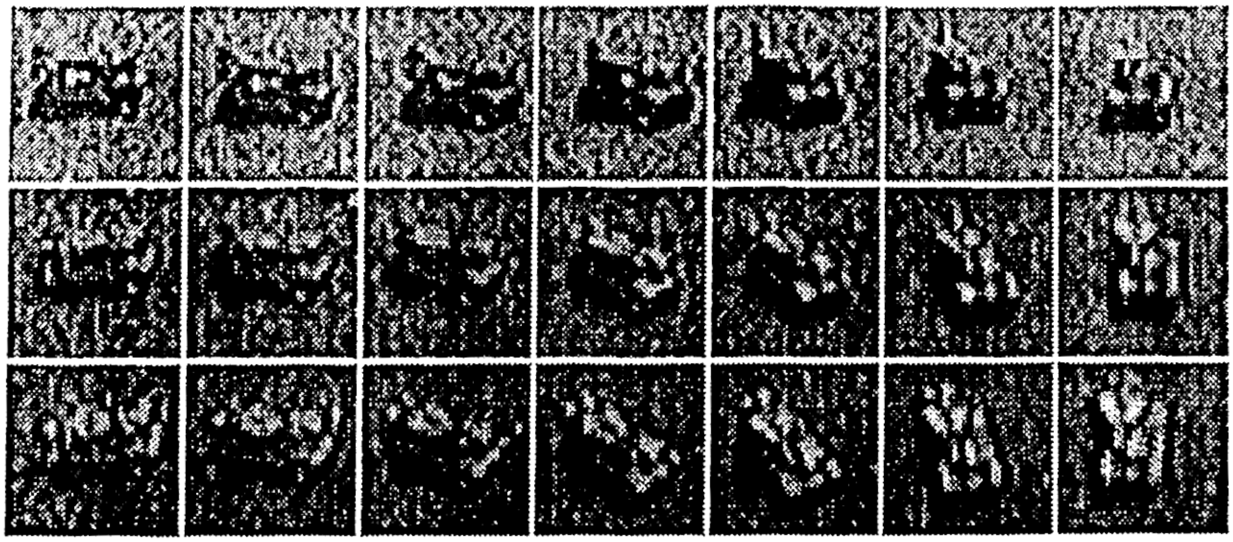
a. Tank

Figure 3. Training images every 15° in both azimuth and elevation. Azimuth runs from 0° to 90° left to right; elevation runs from 15° to 45° top to bottom



b. Armored Personnel Carrier

Figure 3. (continued)



c. Rocket Launcher

Figure 3. (continued)

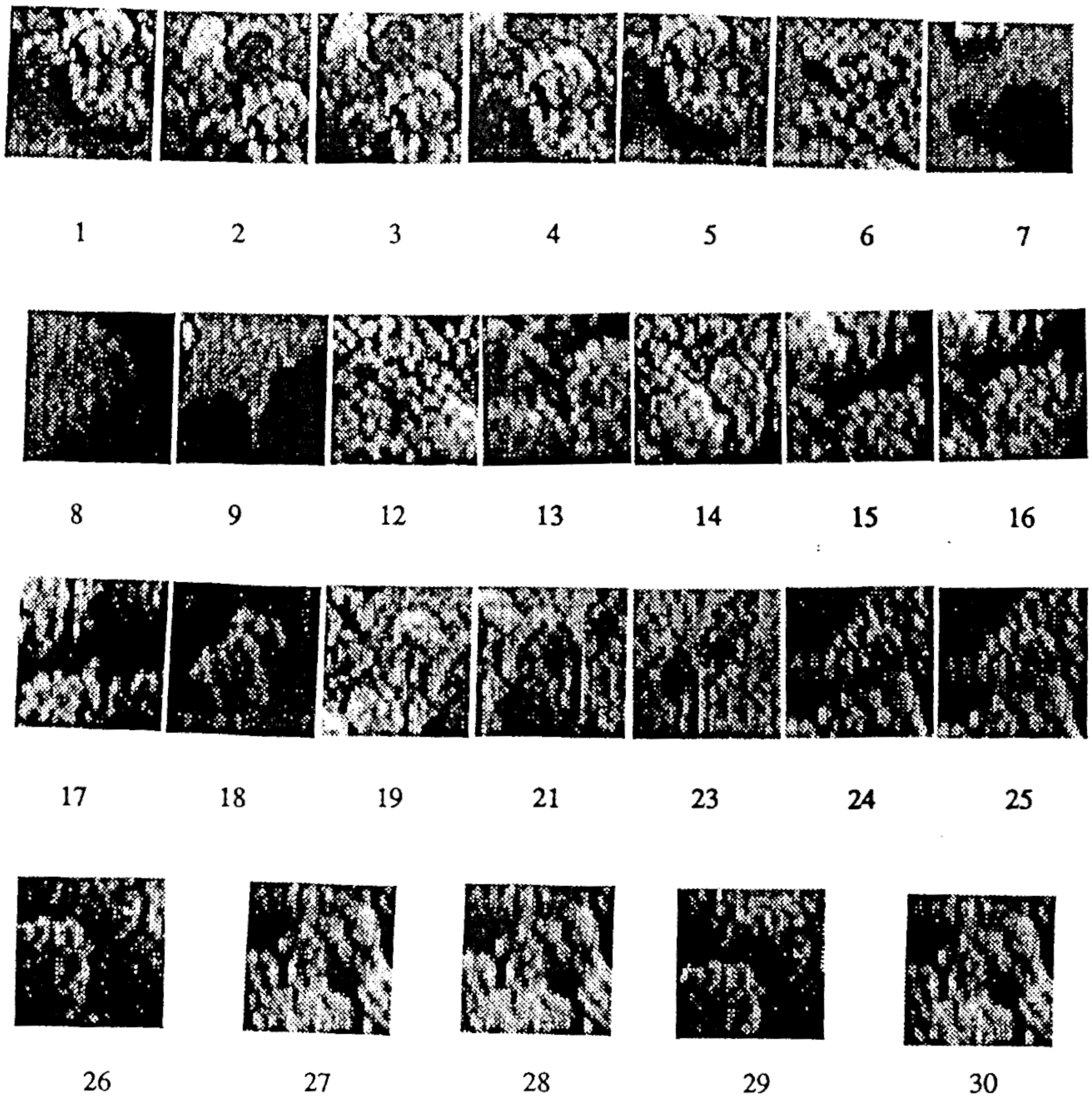
Tank APC Rocket Launcher

a. Target images

#10 (Tank) #11 (Tank) #20 (APC) #22 (APC)

b. Non-target images identified as targets

Figure 4. Test images from terrain board image in Figure 1.



c. Non-target images identified as non-targets

Figure 4. (continued)